

ADOPTING CROSS CHANNEL COMMUNICATION BLOCK in to the GENERATIVE ADVERSARIAL NETWORK for DENOISING of LOW DOSE CT IMAGES

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Abstract

The CT's continued advancement and wide-ranging application in medical operations have prompted questions over the radiation dose connected to patients. Since low-dose CT (LDCT) may reduce the radiation risk, it has drawn a lot of attention. However, merely reducing radiation dosages will cause a conspicuous decline in image quality. Consequently, reducing the noise in the LDCT image is a preferable solution to this problem. Numerous studies have used GANs to significantly improve LDCT image denoising. Current GAN-based LDCT image denoising approaches struggle to provide better structural detail retention. This is mostly due to model training not placing enough emphasis on pixel-wise loss function tolerance and structural feature retention. This study investigates the use of cross-channel communication via an adequate attention mechanism, i.e. efficient channel attention, by extending the recently developed deep learning model known as DU-GAN with dual domain U-Net discriminators, which outperforms the leading models in the field of LDCT restoration. Moreover, a lot of work was put forward to combine the Dice loss function which guides the denoising process in the image domain discriminator, to ensure that the final denoised results are as close as feasible to the industry's gold standard, NDCT. The aforementioned extensions are combined with instance normalization to increase training effectiveness by minimizing the statistical difference between the CT images. The results of our trials using actual clinical images reveal that the proposed method has improved when compared to other experiments and measurements. The suggested model improves the DU-GAN baseline work from 0.72109 to 0.73830, from 20.40010 to 22.04259, and from 0.09821 to 0.08170, respectively, based on SSIM, PSNR, and RMSE. This study finds that integrating effective channel attention into the generator network and utilizing Dice loss as a pixel-wise loss in image domain discriminator instead of Mean Absolute Error loss increase the quality of the denoised LDCT images.

Keywords: DU-GAN, LDCT, NDCT, U-Net, Denoising, Dice loss, over-smoothing, structural detail preservation

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